

Homework 5

Due September 30, 2025.

Problem 1.

Show that in odd dimensions (Euclidean or Minkowskian) both irreps of the Dirac matrices are real or pseudoreal if $C = C_+$, but if $C = C_-$ then the irreps are complex. (*Hint:* For C_+ this is obvious. For C_- show that if $\gamma^{\mu,T} = -C_- \gamma^\mu C_-^{-1}$, then there is no matrix S such that also $\gamma^{\mu,T} = S \gamma^\mu S^{-1}$.)

Problem 2.

Show that one can always rescale C_+ and C_- such that they become unitary matrices. (See the notes for a hint.)

Problem 3.

Complete in table 11.2 on page 153 the entries for the case $6 + 1$.

Comment: These properties will be used when we discuss the spinor irreps of orthogonal Lie algebras.

Problem 4.

Show that if in dimension $d = s + t$ no Majorana spinor exists, a symplectic Majorana spinor (defined in the notes) satisfies the consistency condition for a symplectic Majorana spinor to exist.

Bonus Problem.

When we constructed particular irreps for the Dirac matrices in arbitrary dimensions, we justified this by claiming that many of the properties of Dirac matrices are independent of a particular representation. We now prove this claim. Show that

- (i) if C_+ or C_- is symmetric or antisymmetric, then it remains so after a similarity transformation.
- (ii) the reality properties of an irrep (real, pseudoreal, or complex) do not change by a similarity transformation.

Homework 6

Due October 9, 2025.

Problem 1.

Because it is often needed in string theory, supersymmetry, and supergravity, and also because this is an excellent exercise to prepare for the midterm exam, we study the Dirac matrices and spinors in $d = 3 + 1$, $d = 4 + 0$, $d = 9 + 1$, $d = 10 + 0$, $d = 10 + 1$, $d = 11 + 0$.

- a) Begin with $d = 3 + 1$. What are the reality properties of γ^μ ? Do Majorana or symplectic Majorana spinors exist? Are the charge conjugation matrices symmetric or antisymmetric? Are they block-diagonal or block off-diagonal? What are their spinor indices?
- b) Repeat for $d = 4 + 0$.
- c) Repeat for $d = 9 + 1$.
- d) Repeat for $d = 10 + 0$.
- e) Repeat for $d = 10 + 1$.
- f) Repeat for $d = 11 + 0$.
- g) In which cases do Majorana-Weyl spinors exist, and in which cases do symplectic Majorana-Weyl spinors exist?

Homework 7: Solutions

Problem 1.

Construct the adjoint representation of $SU(2)$ (antihermitian 3×3 matrices) in two ways:

- a) Using the structure constants $f_{ab}{}^c = \epsilon_{abc}$, and
- b) Using raising (J_+) and lowering (J_-) operators. (In this case J_+ , J_- and J_3 generate $SU(1,1)$ which is the Cartan-Weyl basis for $SU(2)$. So you have to transform from hermitian J_+ and J_- to antihermitian T_1 and T_2 .)

Find the similarity transformation S which maps $T_a^{(II)}$ obtained in the second case to the $T_a^{(I)}$ of the first case

$$ST_a^{(II)}S^{-1} = T_a^{(I)}. \quad (1)$$

Solution: In this problem we use some $SU(2)$ technology which we assume to be known. We adopt the following conventions

Compact basis: $[T_a, T_b] = \epsilon_{abc}T_c$; $T_a^\dagger = -T_a$, 3×3 matrices

CW basis: $J_+ = J_1 + iJ_2$, $J_- = J_1 - iJ_2$; $-iJ_a = T_a$ (2)

$$J_+|j, m\rangle = \sqrt{(j+m+1)(j-m)}|j, m+1\rangle; \quad J_-|j, m\rangle = \sqrt{(j+m)(j-m+1)}|j, m-1\rangle.$$

- a) Using $(T_a)^b{}_c = f_{ac}{}^b = \epsilon_{acb}$ (one may check that they satisfy $[T_a, T_b] = \epsilon_{abc}T_c$), we get

$$T_1^{(I)} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}; \quad T_2^{(I)} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}; \quad T_3^{(I)} = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (3)$$

One may check that $[T_1, T_2] = T_3$.

- b) $J_+|1, 1\rangle = 0$, $J_+|1, 0\rangle = \sqrt{2}|1, 1\rangle$, $J_+|1, -1\rangle = \sqrt{2}|1, 0\rangle$. So

$$J_+ = \sqrt{2} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}. \quad (4)$$

Then

$$J_- = J_+^\dagger = \sqrt{2} \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}; \quad J_3 = \frac{1}{2}[J_+, J_-] = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}. \quad (5)$$

(One can also determine J_- and J_3 in the way we did for J_+ .) Hence

$$J_1 = \frac{J_+ + J_-}{2} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}; \quad J_2 = \frac{J_+ - J_-}{2i} = \frac{1}{i\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}; \quad J_3 = \begin{pmatrix} 1 & & \\ & 0 & \\ & & -1 \end{pmatrix}. \quad (6)$$

Then $T_a^{(II)} = -iJ_a^{(II)}$, hence

$$T_1^{(II)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ -i & 0 & -i \\ 0 & -i & 0 \end{pmatrix}; \quad T_2^{(II)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}; \quad T_3^{(II)} = \begin{pmatrix} -i & & \\ & 0 & \\ & & i \end{pmatrix}. \quad (7)$$

We must now find a matrix S such that $ST_a^{(II)} = T_a^{(I)}S$. By brute force (maybe a simpler method exists), we find

$$\begin{aligned} S_{12}(-i) &= 0, & \frac{1}{\sqrt{2}}S_{22}(-i) &= -S_{31}, & S_{11}(-i) + S_{13}(-i) &= 0, & \text{etc.} \\ \Rightarrow S &= \begin{pmatrix} -i & 0 & i \\ 1 & 0 & 1 \\ 0 & i\sqrt{2} & 0 \end{pmatrix}. \end{aligned} \quad (8)$$

As a check we evaluate

$$\begin{aligned} ST_1^{(II)} &= T_1^{(I)}S : \begin{pmatrix} -i & 0 & i \\ 1 & 0 & 1 \\ 0 & i\sqrt{2} & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ -i & 0 & -i \\ 0 & -i & 0 \end{pmatrix} \stackrel{?}{=} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} -i & 0 & i \\ 1 & 0 & 1 \\ 0 & i\sqrt{2} & 0 \end{pmatrix} \\ &= \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 0 & 0 \\ 0 & -2i & 0 \\ \sqrt{2} & 0 & \sqrt{2} \end{pmatrix} \stackrel{?}{=} \begin{pmatrix} 0 & 0 & 0 \\ 0 & -i\sqrt{2} & 0 \\ 1 & 0 & 1 \end{pmatrix} \text{ Agrees.} \end{aligned} \quad (9)$$

Problem 2.

Use the “master formula” of chapter 23 of the notes to construct the representation of $SU(3)$ whose highest weight vector is twice the first fundamental weight vector. Draw the weight diagram. Then draw in this diagram the complex conjugate representation. What is the highest weight of the latter?

Solution: The first fundamental weight vector of $SU(3)$ is $\vec{\mu}_{def}^1 = (\frac{1}{2}, \frac{1}{2\sqrt{3}})$. So we begin with

$$\vec{\mu}_{HW} = (1, \frac{1}{\sqrt{3}}). \quad (10)$$

We use as simple roots of $SU(3)$

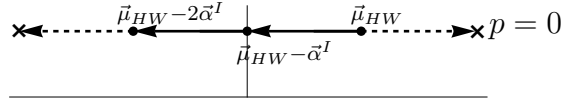
$$\vec{\alpha}^I = (1, 0); \quad \vec{\alpha}^{II} = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right). \quad (11)$$

For any weight $\vec{\mu}_R$ of a representation R the master formula reads

$$2 \frac{\vec{\alpha}^J \cdot \vec{\mu}_R}{\vec{\alpha}^J \cdot \vec{\alpha}^J} = -p + q, \quad \vec{\alpha}^J \text{ are simple roots.} \quad (12)$$

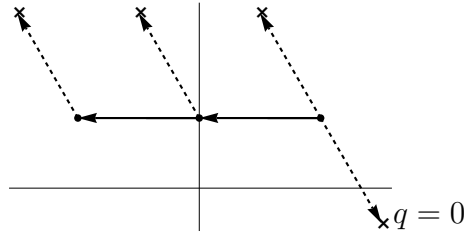
For the highest weight $p = 0$, so we begin with

Step I: $2 \frac{\vec{\alpha}^I \cdot \vec{\mu}_{HW}}{\vec{\alpha}^I \cdot \vec{\alpha}^I} = q = 2$.

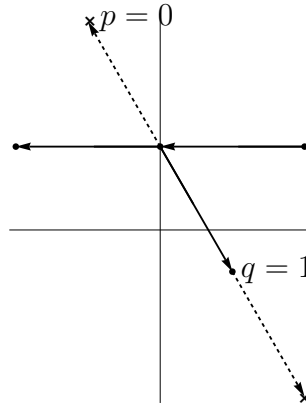


The notation $\cdots \rightarrow \times$ means that there is no weight at $\times$.

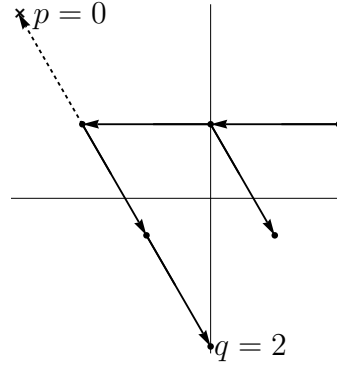
Step II: $2 \frac{\vec{\alpha}^{II} \cdot \vec{\mu}_{HW}}{\vec{\alpha}^{II} \cdot \vec{\alpha}^{II}} = q = 0$.



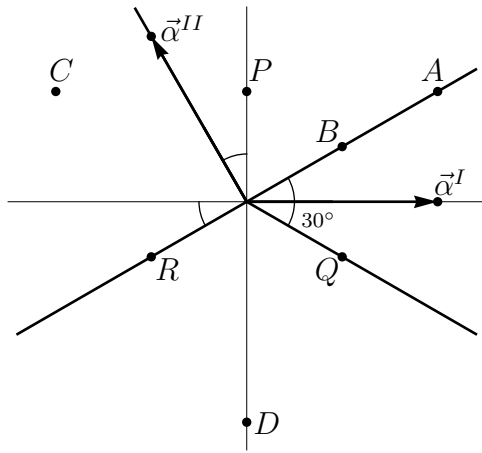
Step III: At $\vec{\mu}_{HW} - \vec{\alpha}^I$ we get $2 \frac{\vec{\alpha}^{II} \cdot (\vec{\mu}_{HW} - \vec{\alpha}^I)}{\vec{\alpha}^{II} \cdot \vec{\alpha}^{II}} = -p + q = q = 1$.



Step IV: At $\vec{\mu}_{HW} - 2\vec{\alpha}^I$ we get $2\frac{\vec{\alpha}^{II} \cdot (\vec{\mu}_{HW} - 2\vec{\alpha}^I)}{\vec{\alpha}^{II} \cdot \vec{\alpha}^{II}} = q = 2$.



We could repeat these steps at all remaining roots, but it is faster to use Weyl reflections. Weyl reflections show that we got all roots.



Weyl reflections:

$$A \rightarrow C \rightarrow D$$

$$P \rightarrow Q \rightarrow R$$

$$\vec{\alpha}^I = (1, 0)$$

$$\vec{\alpha}^{II} = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$$

$$B = \vec{\mu}_{def}^1 = \left(\frac{1}{2}, \frac{1}{2\sqrt{3}}\right)$$

$$A = \left(1, \frac{1}{\sqrt{3}}\right) = 2\vec{\mu}_{def}^1$$

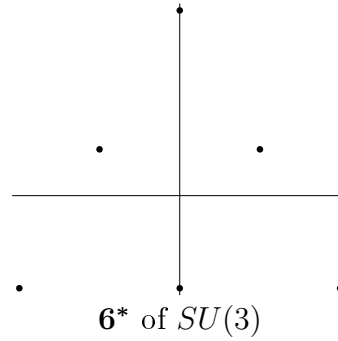
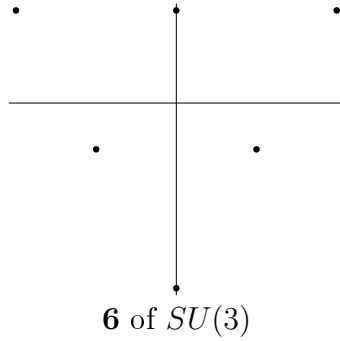
$$C = \left(-1, \frac{1}{\sqrt{3}}\right)$$

$$D = \left(0, -\frac{2}{\sqrt{3}}\right)$$

$$P = \left(0, \frac{1}{\sqrt{3}}\right)$$

$$Q = \left(\frac{1}{2}, -\frac{1}{2\sqrt{3}}\right)$$

$$R = \left(-\frac{1}{2}, -\frac{1}{2\sqrt{3}}\right)$$

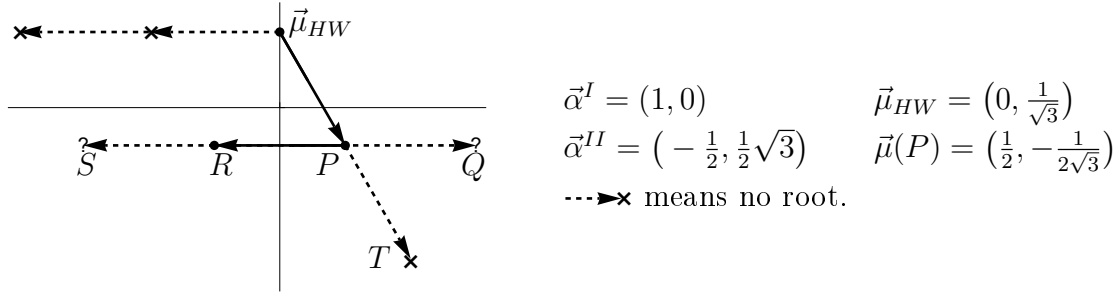


The complex conjugate diagram has as highest weight minus the lowest weight of the **6**, hence

$$\vec{\mu}_{HW}(\mathbf{6}^*) = \left(0, \frac{2}{\sqrt{3}}\right). \quad (13)$$

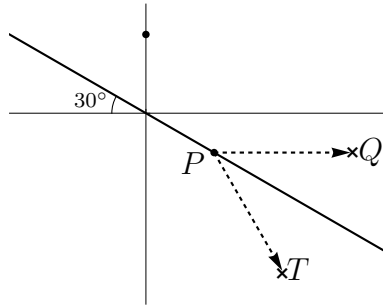
Bonus Problem.

We considered in class the $SU(3)$ representation whose highest weight is $(0, \frac{1}{\sqrt{3}})$. Using the master formula, we came as far as the figure indicates. We found a root at P , not at T .



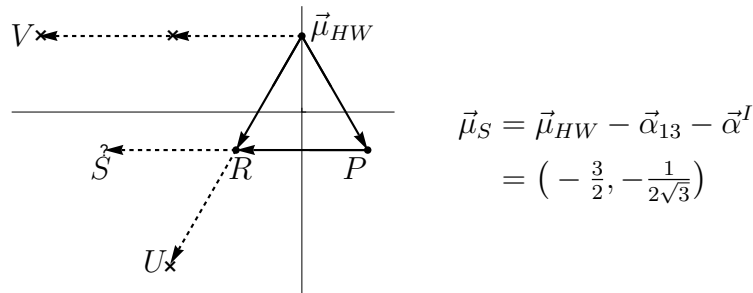
At P one finds from the master formula with $\vec{\alpha}^I$ that $-p + q = 1$. We want to show that $p = 0$ (no root at Q); then we only get a root at R . Prove, using the master formula, that there is no root at Q . Can one conclude this from Weyl reflections?

Solution: Under Weyl reflection Q is mapped to T , but there is no root at T , so there is no root at Q .



One can also prove this from the master equation. If there were a root at Q , there also would be a root at S . But using $\vec{\alpha}^{13}$ (the highest root) in the master formula at $\vec{\mu}_{HW}$ yields $q = 1$.

$$2 \frac{\vec{\alpha}^{13} \cdot \vec{\mu}_{HW}}{\vec{\alpha}^{13} \cdot \vec{\alpha}^{13}} = -p + q = q = 2\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right) \left(0, \frac{1}{\sqrt{3}}\right) = 1. \quad (14)$$



This implies that there is a root at R but no root at U . Applying the master formula at S , we get

$$2 \frac{\vec{\alpha}^{II} \cdot \vec{\mu}_S}{\vec{\alpha}^{II} \cdot \vec{\alpha}^{II}} = -p + q = 2\left(-\frac{3}{2}, -\frac{1}{2\sqrt{3}}\right) \left(-\frac{1}{2}, \frac{1}{2}\sqrt{3}\right) = 2\left(\frac{3}{4} - \frac{1}{4}\right) = 1. \quad (15)$$

But $p = 0$ (no root at V) and $q = 0$ (no root at U). Hence we get a contradiction: $1 = 0$, so there is no root at S .

Homework 7

Due October 28, 2025.

Problem 1.

Construct the adjoint representation of $SU(2)$ (antihermitian 3×3 matrices) in two ways:

- a) Using the structure constants $f_{ab}{}^c = \epsilon_{abc}$, and
- b) Using raising (J_+) and lowering (J_-) operators. (In this case J_+ , J_- and J_3 generate $SU(1,1)$ which is the Cartan-Weyl basis for $SU(2)$. So you have to transform from hermitian J_+ and J_- to antihermitian T_1 and T_2 .)

Find the similarity transformation S which maps $T_a^{(II)}$ obtained in the second case to the $T_a^{(I)}$ of the first case

$$ST_a^{(II)}S^{-1} = T_a^{(I)}. \quad (1)$$

Problem 2.

Use the “master formula” of chapter 23 of the notes to construct the representation of $SU(3)$ whose highest weight vector is twice the first fundamental weight vector. Draw the weight diagram. Then draw in this diagram the complex conjugate representation. What is the highest weight of the latter?

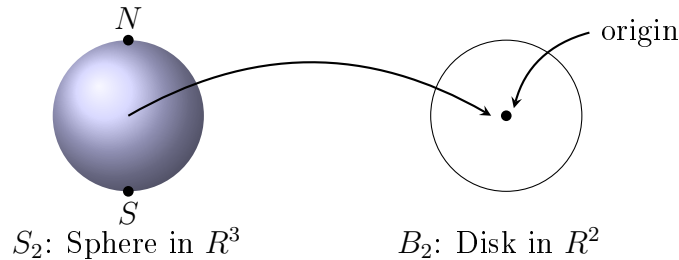
Homework 8: Solutions

Problem 1.

We showed in class that S_3 and B_3 are topologically equivalent (with all points of the boundary of B_3 identified as only point), namely there is a 1-1 map between them which is continuous.

- a) Now analyse the case S_2 and B_2 .

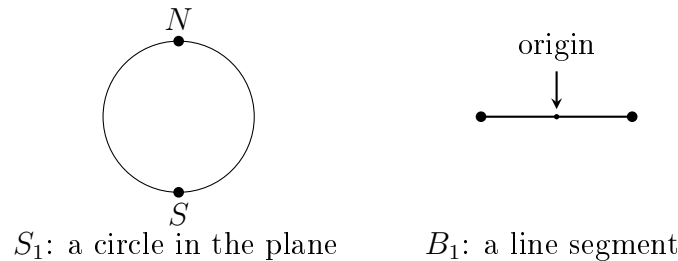
Solution:



We wrap the disk around the sphere: origin of disk at North pole of sphere, the whole boundary of the disk becomes the South pole. So all points of the circle that is the boundary of B_2 are identified.

- b) Next analyse the case S_1 and B_1 .

Solution:



Wrap the line segment around the circle. The two endpoints of the line segment must be identified because they become the South pole.

- c) In which of these 3 cases is the sphere isomorphic to a group, and which group?

Solution: S_3 is isomorphic to $SU(2)$. S_2 is not a group (the LA with two generators is abelian, namely two commuting generators A and B). S_1 is the “circle group”, namely $U(1)$.

- d) Which of these groups is a covering group, and which of them is not simply-connected?

Solution: $SU(2)$ is a covering group (of $SO(3)$) and is simply-connected. S_1 is not simply connected, so it is **not** a covering group. S_1 and B_1 are topologically the same, their covering group is R^1 (unwind the circle, as discussed in class).

Problem 2.

We proved that $SU(2)/Z_2 \simeq SO(3)$ by considering the equation

$$u^{-1}x^k\sigma_k u = x'^k\sigma_k = R^i_j x^j\sigma_i, \quad u = e^{-\frac{i}{2}\omega^k\sigma_k}. \quad (1)$$

- a) Prove that the left-hand side is linear in σ_l (expand $e^{-\frac{i}{2}\omega^k\sigma_k}$) and determine R^i_j .

Solution: We have to show that expanding u we only get terms with σ_k but no terms with I .

$$\begin{aligned} \left(\cos \frac{\omega}{2} + i \frac{\omega^l}{\omega} \sin \frac{\omega}{2} \sigma_l \right) \sigma_k \left(\cos \frac{\omega}{2} - i \frac{\omega^m}{\omega} \sin \frac{\omega}{2} \sigma_m \right) \\ = \cos^2 \frac{\omega}{2} \sigma_k + \left[i \frac{\omega^l}{\omega} \sin \frac{\omega}{2} \sigma_l, \sigma_k \right] \cos \frac{\omega}{2} + \frac{\omega^l}{\omega} \frac{\omega^m}{\omega} \sin^2 \frac{\omega}{2} \sigma_l \sigma_k \sigma_m \end{aligned} \quad (2)$$

Using $\omega^l \omega^m \sigma_l \sigma_k \sigma_m = \omega_k \omega^m \sigma_m + \omega^l \sigma_l \omega_k - \omega^2 \sigma_k$ we get

$$\begin{aligned} &= \left(\cos^2 \frac{\omega}{2} - \sin^2 \frac{\omega}{2} \right) \sigma_k + i \frac{\omega^l}{\omega} \sin \frac{\omega}{2} \cos \frac{\omega}{2} (2i \epsilon_{lkm} \sigma_m) + 2 \frac{\omega^k \omega^l}{\omega^2} \sigma_l \sin^2 \frac{\omega}{2} \\ &= \cos \omega \sigma_k + \epsilon_{klm} \frac{\omega^l}{\omega} \sigma_m \sin \omega + \frac{\omega^k \omega^l}{\omega^2} \sigma_l (1 - \cos \omega) \\ &= \left(\delta^{kl} - \frac{\omega^k \omega^l}{\omega^2} \right) \cos \omega \sigma_l + \frac{\omega^k \omega^l}{\omega^2} \sigma_l + \epsilon_{klm} \frac{\omega^l}{\omega} \sin \omega \sigma^m \equiv R^k_l \sigma^l \\ \Rightarrow R^k_l &= \left(\delta^k_l - \frac{\omega^k \omega^l}{\omega^2} \right) \cos \omega + \frac{\omega^k \omega^l}{\omega^2} - \epsilon_{klm} \frac{\omega^m}{\omega} \sin \omega. \end{aligned} \quad (3)$$

- b) Prove that R is a rotation (orthogonal with $\det R = 1$) and does not rotate the axis of rotation.

Hint: Consider $\text{tr } x \cdot \sigma x \cdot \sigma$ and $\text{tr } x' \cdot \sigma x' \cdot \sigma$.

Solution: $\text{tr } x \cdot \sigma x \cdot \sigma = x^i x^j \text{tr } \sigma_i \sigma_j = 2\vec{x}^2$ and $\text{tr } x' \cdot \sigma x' \cdot \sigma = \text{tr}(u^{-1}x \cdot \sigma u)(u^{-1}x \cdot \sigma u) = \text{tr } x \cdot \sigma x \cdot \sigma = 2(R^i_j x^j)^2$, hence $\vec{x}^2 = (\vec{x}')^2$. So R preserves lengths. Then $x^T R^T R x = x^T x$, and by considering $x + \alpha y$, we find $x^T R^T R y = x^T y$. Then $R^T R = I$, thus R is orthogonal. Then $\det R = \pm 1$, but at $\omega = 0 \Rightarrow u = 1$ we get $R = I$, hence $\det R = 1$. Thus R lies in $SO(3)$. And $R^k_l \omega^l = \omega^k$: R does not rotate the axis of rotation $\vec{\omega}$.

- c) Consider the homomorphism $\phi(u) = R$ in a). What is the kernel?

Solution: The kernel of $\phi(u) = R$ are those group elements of $SU(2)$ for which $\phi(u) = I$. So $u^{-1}\sigma_k u = \sigma_k$. We already calculated $u^{-1}\sigma_k u$; it gave $R^k_l \sigma_l$. So we must find those $\vec{\omega}$ for which $R^k_l = \delta^k_l$. That is the case if $\cos \omega = 1$ (in which case $\sin \omega = 0$). Then $\omega = 0$ or $\omega = 2\pi$, which gives $u = I$ and $u = -I$. So the kernel is $(I, -I)$ which forms the group Z_2 .

Problem 3. The Lorentz group $L = O(3, 1)$.

The group $O(3, 1)$ preserves $x^\mu \eta_{\mu\nu} x^\nu$ with $\eta_{\mu\nu} = (-1, +1, +1, +1)$, hence $x'^\mu = L^\mu{}_\nu x^\nu$ with $L^T \eta L = \eta$.

a) Show that $\det L = \pm 1$.

Solution: If $L^T \eta L = \eta$, then $(\det L)^2 = 1$, so $\det L = \pm 1$.

b) Show that $L^0{}_0$ is either ≥ 1 or ≤ -1 .

Solution: From $L^T \eta L = \eta$ or $(L^T)^\nu{}_\mu \eta_{\nu\rho} L^\rho{}_\sigma = \eta_{\mu\sigma}$ we get for $\mu = \sigma = 0$

$$\begin{aligned} L^\nu{}_0 \eta_{\nu\rho} L^\rho{}_0 &= \eta_{00} = -1 \\ -(L^0{}_0)^2 + \sum_{j=1}^3 (L^j{}_0)^2 &= -1. \end{aligned} \quad (4)$$

Hence $(L^0{}_0)^2 = 1 + \sum (L^j{}_0)^2 \geq 1$. Then $|L^0{}_0| \geq 1$, or $L^0{}_0 \geq 1$ or $L^0{}_0 \leq -1$.

c) So $O(3, 1)$ contains 4 disconnected components. Write

$$O(3, 1) = SO(3, 1)^\uparrow \cup \Lambda_P SO(3, 1)^\uparrow \cup \Lambda_T SO(3, 1)^\uparrow \cup \Lambda_{PT} SO(3, 1)^\uparrow. \quad (5)$$

Find diagonal matrices for Λ_P , Λ_T and Λ_{PT} .

Solution: The four sets of group elements with $\det L = \pm 1$ and $L^0{}_0 \geq 1$ or ≤ -1 cannot be connected by a continuous curve. (One cannot continuously transform from $\det L = +1$ to $\det L = -1$, or from $L^0{}_0 \geq 1$ to $L^0{}_0 \leq -1$). Any group element g in one of these set can be written as a coset representative g_R times a group element with $\det L = 1$ and $L^0{}_0 \geq 1$, because $g = g_R(g_R^{-1}g)$ and $g_R^{-1}g$ has $\det(g_R^{-1}g) = 1$ and $L^0{}_0(g_R^{-1}g) \geq 1$. The latter follows from $L^{-1} = \eta L^T \eta$ which implies $(L^{-1})^0{}_0 = L^0{}_0$ and $(L^{-1})^0{}_j = -L^j{}_0$. Then $L(g_R^{-1}g)^0{}_0$ is equal to $L(g_R^{-1})^0{}_0 L(g)^0{}_0 + L(g_R^{-1})^0{}_j L(g)^j{}_0 = L(g_R)^0{}_0 L(g)^0{}_0 - L(g_R)^0{}_j L(g)^j{}_0$ and the triangle inequality proves that the first term dominates. As coset representatives we choose

$$\Lambda_P = \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix}, \quad \Lambda_T = \begin{pmatrix} -1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{pmatrix}, \quad \Lambda_{PT} = \begin{pmatrix} -1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix}. \quad (6)$$

d) Consider the homomorphism $\phi : L \rightarrow (\det L, \text{sign } L^0{}_0)$. Which group V do the four group elements $(\det L, \Lambda^0{}_0) = (\pm 1, \pm 1)$ form?

Solution: The set of 4 group elements $e = (1, 1)$, $a = (1, -1)$, $b = (-1, 1)$ and $c = (-1, -1)$ forms the Klein group $V = (e, a, b, c)$ because $ab = c$, $ac = b$, $bc = a$, $a^2 = b^2 = c^2 = 1$ and it is abelian.

- e) Now show that $SO(3,1)^\uparrow$ is the kernel of the homomorphism $O(3,1) \rightarrow V$. Is $SO(3,1)^\uparrow$ a normal subgroup of $O(3,1)$?

Solution: The map ϕ which maps any group element L of $O(3,1)$ to a group element $(\det L, \text{sign } L^0_0)$ is a homomorphism because

$$\det L_1 L_2 = \det L_1 \det L_2 \text{ and } \text{sign } (L_1 L_2)^0_0 = \text{sign } (L_1)^0_0 \text{sign } (L_2)^0_0. \quad (7)$$

The latter we proved in c). The kernel consists of those L with $\det L = +1$ and $L^0_0 \geq 1$. That is $SO(3,1)^\uparrow$. Since $SO(3,1)^\uparrow$ is the kernel, it is a normal subgroup of $O(3,1)$.

Problem 4.

We want to prove that $Sl(2, \mathbb{C})$ is the covering group of $SO(3,1)^\uparrow$. Consider the expression $Ax^\mu \sigma_\mu A^\dagger$ with A in $Sl(2, \mathbb{C})$ and $\sigma_\mu = (-I, \sigma_1, \sigma_2, \sigma_3)$. (Note $A^\dagger$, not A^{-1} this time.)

- a) Show that this expression is equal to $x'^\mu \sigma_\mu$, where $x'^\mu = L^\mu_\nu x^\nu$ with **real** L^μ_ν .

Hint: Can one always write a complex 2×2 matrix as $x'^\mu \sigma_\mu$?

Solution: $Ax^\mu \sigma_\mu A^\dagger$ is a 2×2 matrix. Any 2×2 matrix can be expanded on σ^k, I . So $Ax^\mu \sigma_\mu A^\dagger = x'^\mu \sigma_\mu$. However, $Ax^\mu \sigma_\mu A^\dagger$ is linear in x^ν so x'^μ is linear in x^ν . Then $x'^\mu = L^\mu_\nu x^\nu$. Since $Ax^\mu \sigma_\mu A^\dagger$ is hermitian, x'^μ is real. So L^μ_ν is real.

- b) Show that $\det x^\mu \sigma_\mu = \det x'^\mu \sigma_\mu$, and then show that $L^\mu_\nu \in O(3,1)$.

Solution: $\det Ax^\mu \sigma_\mu A^\dagger = \det x^\mu \sigma_\mu$ because $\det A = \det A^\dagger = 1$. Then

$$\det x^\mu \sigma_\mu = \det \begin{pmatrix} -x^0 + x^3 & x^1 - ix^2 \\ x^1 + ix^2 & -x^0 - x^3 \end{pmatrix} = (x^0)^2 - (x^3)^2 - (x^1)^2 - (x^2)^2 \equiv -x^\mu \eta_{\mu\nu} x^\nu. \quad (8)$$

Similarly $\det x'^\mu \sigma_\mu = -x'^\mu \eta_{\mu\nu} x'^\nu = -(L^\mu_\rho x^\rho) \eta_{\mu\nu} (L^\nu_\sigma x^\sigma) = -x^\rho (L^T)_\rho^\mu \eta_{\mu\nu} L^\nu_\sigma x^\sigma = -x^T L^T \eta L x$. So $x^T \eta x = x^T L^T \eta L x$. Replacing x by $x + \alpha y$ we obtain (done in class for $SO(N)$)

$$x^T \eta y = x^T L^T \eta L y \quad \Rightarrow \quad L^T \eta L = \eta. \quad (9)$$

- c) Now show that $L^\mu_\nu \in SO(3,1)^\uparrow$. We denote these group elements by $\hat{L}^\mu_\nu$.

Solution: From (9) it follows that L lies in the group $O(3,1)$. But for $A \rightarrow I$ we get $x'^\mu = x^\mu$, so $L = I$. So L is connected to the unit I . So $\det L = +1$ and $L^0_0 \geq 1$. Thus L lies in $SO(3,1)^\uparrow$.

- d) Show that the map $\phi(A) = \hat{L}$ is a homomorphism. What is the kernel of ϕ ?

Solution: $BA\sigma_\mu x^\mu A^\dagger B^\dagger = B(A\sigma_\mu x^\mu A^\dagger)B^\dagger = B(x'^\mu \sigma_\mu)B^{-1} = L_B^\mu_\nu x'^\nu \sigma_\mu = L_B^\mu_\nu L_A^\nu_\rho x^\rho \sigma_\mu$ since $x'^\nu = L_A^\nu_\rho x^\rho$. On the other hand $BA\sigma_\mu x^\mu A^\dagger B^\dagger = (BA)\sigma_\mu x^\mu (BA)^\dagger = x'^\mu \sigma_\mu$ with

$x'^\mu = L_{BA}{}^\mu{}_\nu x^\nu$. So $L_B L_A = L_{BA}$. Then $\phi(BA) = L_{BA} = L_B L_A = \phi(B)\phi(A)$. so $\phi(A) = \hat{L}$ is a homomorphism. The kernel of ϕ are all matrices A such that $Ax^\mu \sigma_\mu A^\dagger = x^\mu \sigma_\mu$. Two solutions are obvious: $A = I$ and $A = -I$. To show that there are no more we set $A = aI + b^k \sigma_k$ with complex a and b^k satisfying $a^2 - \sum_{k=0}^3 (b^k)^2 = 1$. Then we must solve

$$(aI + b^k \sigma_k) \sigma_\mu (a^* I + (b^l)^* \sigma_l) = \sigma_\mu. \quad (10)$$

For $\sigma_\mu = I$ we get $|a|^2 + \sum |b^k|^2 = 1$ and $\epsilon_{klm} b^k (b^l)^* = 0$. Clearly $a^2 = |a|^2 = 1$, so $a = \pm 1$.

e) Show that $Sl(2, \mathbb{C})/Z_2 \simeq SO(3, 1)^\dagger$.

Solution: Since $Z_2 = (I, -I)$ is the kernel of the homomorphism of $SU(2)$ into $SO(3)$ we have $Sl(2, \mathbb{C})/Z_2 = SO(3, 1)$ if we can prove that the homomorphism is not only “into” but also “onto”. Given an A in $Sl(2, \mathbb{C})$ we always get an L in $SO(3, 1)$. All L can be obtained from the pure boosts and pure rotations. The matrices $e^{\vec{\omega} \cdot \vec{\sigma}}$ give all boosts and the matrices $A = e^{i\vec{\omega} \cdot \vec{\sigma}}$ give all rotations. Hence we get the whole $SO(3, 1)$.

Bonus Problem.

Show that $Sl(2, \mathbb{C})$ is simply connected.

a) First assume that $A \in Sl(2, \mathbb{C})$ has the polar decomposition $A = HU$ where $U \in SU(2)$ and H is a 2×2 matrix which is hermitian, has $\det H = 1$, and $\text{tr } H > 0$. Show that the manifold of H is a connected component of a hyperboloid. Then use this to show that $Sl(2, \mathbb{C})$ is simply connected.

Solution: If $A = HU$ with A in $Sl(2, \mathbb{C})$ and $U \in SU(2)$ and H hermitian, then $\det H = 1$. We get $AA^\dagger = HH$. Since $AA^\dagger$ is hermitian, it can be decomposed by a unitary matrix V , $AA^\dagger = V^{-1} \tilde{D} V$ with $\tilde{D}$ diagonal. Since $AA^\dagger > 0$ (meaning $x^\dagger AA^\dagger x > 0$ for all x) $\tilde{D}$ has diagonal entries > 0 . Then we can take the square root D of $\tilde{D}$ by just taking the square root of the diagonal entries. So $\tilde{D} = D^2$ and $D > 0$. Then $AA^\dagger = (V^{-1} D V)(V^{-1} D V)$ so we define H by $V^{-1} D V = H$. So we constructed H from A . This H indeed satisfies $\det H = 1$, $H^\dagger = H$, and $\text{tr } H > 0$ (because $D = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$ with $\lambda_1 > 0$ and $\lambda_2 > 0$). Finally, to satisfy $A = HU$, we define U by $U = H^{-1}A$. We must prove that U lies in $SU(2)$: $\det U = 1$ and $UU^\dagger = I$. We already showed that $\det H = \det A = 1$, so we only must show that $UU^\dagger = I$

$$\begin{aligned} UU^\dagger &= H^{-1} AA^\dagger H^{-1} = (V^{-1} D^{-1} V) AA^\dagger (V^{-1} D^{-1} V) \\ &= (V^{-1} D^{-1} V) [(V^{-1} D V)(V^{-1} D V)] (V^{-1} D^{-1} V) \\ &= V^{-1} D^{-1} D D D^{-1} V = I. \end{aligned} \quad (11)$$

b) Show that the matrices H do not form a group (if group multiplication is matrix multiplication).

Solution: The product of two hermitian matrices is in general not hermitian.

c) Now prove the polar decomposition $A = HU$.

Hint: First diagonalize $AA^\dagger$ by a unitary matrix V such that $VAA^\dagger V^{-1} = D^2$. Let D be the positive square root of D^2 , so $\text{tr } D > 0$. Then define $H = V^{-1}DV$.

Solution: See a).

d) Show that $U = H^{-1}A$ lies in $SU(2)$.

Solution: See a).

e) We showed above that $Sl(2, \mathbb{C})$ is simply connected. Is $Gl(2, \mathbb{C})$ also simply connected?

Solution: Consider the map $\phi(g) = \det g$ for g in $Gl(2, \mathbb{C})$. The whole complex plane except the origin is covered by $\det g$. Consider a closed curve around the origin in the complex plane. It corresponds to a closed curve in $Gl(2, \mathbb{C})$. The contraction of this curve yields a point where $\det g = 0$, but $\det g \neq 0$ for groups. Thus $Gl(2, \mathbb{C})$ is not simply connected.

A proof that follows more what we did for $SU(2)/Z_2$ is as follows: Begin with $Gl(N, \mathbb{C})$ and write g in $Gl(N, \mathbb{C})$ as $g_0 z$ where g_0 lies in $Sl(N, \mathbb{C})$ and $z \neq 0$ is a complex number. The decomposition of g into g_0 and z is not unique: one gets the same g if one replaces g_0 by $g_0 e^{2\pi i k/N}$ (which still lies in $Sl(N, \mathbb{C})$) and replaces z by $z e^{-2\pi i k/N}$ which is still a complex number different from 0. The manifold of these points is $C \setminus 0$, the complex plane with the origin removed. Hence we divide by the group elements $e^{2\pi i k/N}$ which form the group Z_N

$$Gl(N, \mathbb{C}) = \frac{Sl(N, \mathbb{C}) \times C \setminus 0}{Z_N}. \quad (12)$$

Now $N = 2$. The group $Sl(2, \mathbb{C})$ is simply connected and the discrete group Z_2 is also simply connected, but $C \setminus 0$ is infinitely connected. Hence $Gl(2, \mathbb{C})$ is not simply connected.

Homework 9: Solutions

Problem 1.

Evaluate the Clebsch-Gordan (CG) decomposition of the product of two adjoint matrices of $SU(3)$. Show that there are two adjoint matrices in this decomposition. Using that CG coefficients are invariant tensors, explain what the relation is between the two adjoint matrices in the decomposition, and the structure constants and the d -symbol. Could there be 3 adjoint matrices in the CG expansion of two adjoint matrices of $SU(N)$? One adjoint matrix? Show that for $SU(N)$ there are again always two adjoint matrices in the expansion.

Solution:

$$\begin{array}{c}
 \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} \otimes \begin{array}{|c|c|} \hline a & a \\ \hline b & \square \\ \hline \end{array} = \left(\begin{array}{|c|c|c|c|} \hline \square & \square & a & a \\ \hline \square & \square & \square & \square \\ \hline \end{array} \oplus \begin{array}{|c|c|c|} \hline \square & \square & a \\ \hline \square & \square & \square \\ \hline \end{array} \oplus \begin{array}{|c|c|c|} \hline \square & \square & a \\ \hline \square & \square & \square \\ \hline \end{array} \oplus \begin{array}{|c|c|c|} \hline \square & \square & a \\ \hline \square & \square & \square \\ \hline \end{array} \right) \otimes \begin{array}{|c|} \hline b \\ \hline \end{array} \\
 \mathbf{8} \quad \times \quad \mathbf{8} \\
 \\
 = \begin{array}{|c|c|c|c|} \hline \square & \square & a & a \\ \hline \square & \square & b & \square \\ \hline \end{array} \oplus \begin{array}{|c|c|c|} \hline \square & \square & a \\ \hline \square & \square & b \\ \hline \end{array} \oplus \begin{array}{|c|c|c|} \hline \square & \square & a \\ \hline \square & \square & b \\ \hline \end{array} \oplus \begin{array}{|c|c|c|} \hline \square & \square & a \\ \hline \square & \square & b \\ \hline \end{array} \oplus \begin{array}{|c|c|c|c|} \hline \square & \square & a & a \\ \hline \square & \square & \square & \square \\ \hline \end{array} \oplus \begin{array}{|c|c|c|} \hline \square & \square & a \\ \hline \square & \square & b \\ \hline \end{array} \\
 \mathbf{27} \quad + \quad \mathbf{10^*} \quad + \quad \mathbf{8} \quad + \quad \mathbf{1} \quad + \quad \mathbf{10} \quad + \quad \mathbf{8}
 \end{array}$$

With tensors:

$$\begin{array}{ccccccc}
 u_{\beta}^{\alpha} v_{\delta}^{\gamma} & = & (uv)_{(\beta\delta)}^{(\alpha\gamma)} & + & (uv)_{[\beta\delta]}^{(\alpha\gamma)} & + & (uv)_{(\beta\delta)}^{[\alpha\gamma]} & + & (uv)_{[\beta\delta]}^{[\alpha\gamma]} & + & (u \cdot v)_{\beta}^{\gamma} & + & u : v \\
 6 \times 6 - 9 = & & 18 - 8 = & & 18 - 8 = & & 8 & & 8 & & 8 & & 1 \\
 \mathbf{27} & & \mathbf{10} & & \mathbf{10^*} & & \mathbf{8} & & \mathbf{8} & & \mathbf{1} & &
 \end{array}$$

The term $(u \cdot v)_{\delta}^{\alpha}$ has the same structure as $(u \cdot v)_{\gamma}^{\beta}$, so we omitted it.

If one couples two octets, and if in the CG decomposition another octet appears, then the CG coupling constant for $8 \times 8 = 8$ is an invariant tensor t_{abc} . There are two such tensors in $SU(N)$: the structure constants f_{abc} ($= f_{ab}{}^{c'} g_{c'c}$) and the d -symbol d_{abc} . One can begin with the symmetric product $(8 \times 8)_s$, and then $t_{abc} = d_{abc}$. On the other hand, $(8 \times 8)_a$ yields f_{abc} . Since there are not 3 invariant tensors t_{abc} , one cannot obtain 3 adjoints in adjoint $\times$ adjoint. One can always project with f_{abc} or d_{abc} on adjoint $\times$ adjoint, so one should always find two adjoints in adjoint $\times$ adjoint for $SU(N)$. Direct evaluation of Young tableaux confirms this:

$$(N-1) \left\{ \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} \right\} \otimes (N-1) \left\{ \begin{array}{|c|c|} \hline a & a \\ \hline b & \square \\ \hline c & \square \\ \hline \vdots & \square \\ \hline e & \square \\ \hline \end{array} \right\} = \left(\begin{array}{|c|c|c|c|} \hline \square & \square & a & a \\ \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \end{array} \oplus (N-1) \left\{ \begin{array}{|c|c|c|} \hline \square & \square & a \\ \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \end{array} \oplus N \left\{ \begin{array}{|c|c|c|} \hline \square & \square & a \\ \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \end{array} \oplus N \left\{ \begin{array}{|c|c|c|} \hline \square & \square & a \\ \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \end{array} \right\} \otimes \left\{ \begin{array}{|c|} \hline b \\ \hline c \\ \hline \vdots \\ \hline e \\ \hline \end{array} \right\} \right\} (N-2)$$

$$= \dots \oplus \begin{array}{|c|c|c|} \hline & & a \\ \hline & a & \\ \hline & b & \\ \hline & c & \\ \hline & & \\ \hline e & & \\ \hline \end{array} \oplus \begin{array}{|c|c|c|} \hline & & a \\ \hline & b & \\ \hline & c & \\ \hline & & \\ \hline & e & \\ \hline a & & \\ \hline \end{array} \quad (1)$$

Using $\#a \geq \#b \geq \#c$, there are only two tableaux of the form

$$N \left\{ \begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline & & \\ \hline & & \\ \hline & & \\ \hline & & \\ \hline & & \\ \hline & & \\ \hline \end{array} \right\} = N - 1 \left\{ \begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline & & \\ \hline & & \\ \hline & & \\ \hline & & \\ \hline & & \\ \hline & & \\ \hline \end{array} \right\}. \quad (2)$$

Problem 2.

Write down the CG decomposition of the product of two spinor irreps of $SO(10)$, and of the product of one spinor irrep and one conjugate spinor irrep. First construct the charge conjugation matrix in $d = 10$, and determine if it is block-diagonal or block-off-diagonal. Are the spinor and conjugate spinor irreps in $d = 10$ real, pseudoreal, or complex?

Solution: We begin by fixing the Dirac matrices in $d = 10 + 0$. We start from $d = 8 + 0$, and go via $d = 9 + 0$ to $d = 10 + 0$.

$\gamma^{(8)}$: 8 real symmetric 16×16 block-off-diagonal matrices. $\gamma_c^{(8)} = \gamma^1 \cdots \gamma^8$ satisfies $(\gamma_c^{(8)})^2 = I$, so $\gamma_c^8 = \gamma^9$ is also real and symmetric, but block-diagonal.

$\gamma^{(9)}$: $\{\gamma^{(8)}, \gamma_c^{(8)}\} = 9$ real symmetric 16×16 matrices.

$\gamma^{(10)}$: $\{\gamma^{(9)} \times \tau_1, I \times \tau_2\}$: 9 real symmetric matrices and one imaginary antisymmetric matrix, all in $d = 10 + 0$. All 10 matrices are hermitian and block-off-diagonal

$$\gamma^{(10)} = \left\{ \begin{pmatrix} 0 & \gamma^{(9)} \\ \gamma^{(9)} & 0 \end{pmatrix}, \begin{pmatrix} 0 & -iI \\ iI & 0 \end{pmatrix} \right\} \quad (3)$$

The charge conjugation matrices C_+ and C_- are as follows

$$\begin{aligned} C_+ \gamma^{(10)} &= \gamma^{10,T} C_+ = (\gamma^9 \times \tau_1, -I \times \tau_2) C_+ \Rightarrow C_+ = I \times \tau_1 \\ C_- \gamma^{(10)} &= -\gamma^{10,T} C_- = (-\gamma^9 \times \tau_1, I \times \tau_2) C_- \Rightarrow C_- = I \times (-i\tau_2) \end{aligned} \quad (4)$$

(One can also set $C_- = I \times \tau_2$, but $I \times (-i\tau_2)$ is real.) The $d = 10$ chirality matrix $\gamma_c^{(10)}$ is given by

$$\gamma_c^{(10)} \sim (\gamma^1 \times \tau_1)(\gamma^2 \times \tau_1) \cdots (\gamma^9 \times \tau_1)(I \times \tau_2) = I \times \tau_1 \tau_2 = I \times i\tau_3 \quad (5)$$

So $\gamma_c^{(10)} = I \times \tau_3$. Then $C_+ = C_- \gamma_c^{(10)}$ is a check.

$$C_+ = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}; \quad C_- = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad (6)$$

For the product of two spinors ψ^α and χ^α (with $\alpha = 1, \dots, 32$, but $\chi^\alpha = \begin{pmatrix} \chi_0^a \\ 0 \end{pmatrix}$ and $\psi^\alpha = \begin{pmatrix} \psi_0^a \\ 0 \end{pmatrix}$ with $a = 1, \dots, 16$) we need an odd number of Dirac matrices because $C_\pm$ and $\gamma^{(10)}$ are block-offdiagonal. This gives

$$\psi^\alpha \chi^\beta = \sum_{k=1,3,5} c_k (\gamma^{\mu_1 \dots \mu_k} C^{-1})^{\alpha\beta} (\chi^T C \gamma_{\mu_1 \dots \mu_k} \psi), \quad (7)$$

where c_k are constants. The spinor space of $\psi^\alpha \chi^\beta$ is $16 \times 16 = 256$ dimensional, and the Dirac matrices on the right-hand side yield a basis with $10 + \left(\frac{10 \cdot 9 \cdot 8}{1 \cdot 2 \cdot 3} = 120\right) + \left(\frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5} \frac{1}{2} = 126\right) = 256$ terms. The factor $\frac{1}{2}$ in the last term is due to the fact that an antisymmetric tensor $t^{\mu_1 \dots \mu_5}$ of $SO(10)$ is reducible: it splits into two irreducible reps

$$\begin{aligned} t^{\mu_1 \dots \mu_5} &= u^{\mu_1 \dots \mu_5} + v^{\mu_1 \dots \mu_5} \\ u^{\mu_1 \dots \mu_5} &= \frac{1}{2} \left(t^{\mu_1 \dots \mu_5} + \frac{i}{5!} \epsilon^{\mu_1 \dots \mu_5 \mu_6 \dots \mu_{10}} t_{\mu_6 \dots \mu_{10}} \right) \\ v^{\mu_1 \dots \mu_5} &= \frac{1}{2} \left(t^{\mu_1 \dots \mu_5} - \frac{i}{5!} \epsilon^{\mu_1 \dots \mu_5 \mu_6 \dots \mu_{10}} t_{\mu_6 \dots \mu_{10}} \right) \end{aligned} \quad (8)$$

For example

$$\begin{aligned} u^{12345} &= \frac{1}{2} (t^{12345} + i t^{678910}) \\ u^{678910} &= \frac{1}{2} (t^{678910} - i t^{12345}) = -i(u^{12345}) \end{aligned} \quad (9)$$

(The $-$ sign is due to $\epsilon^{\mu_1 \dots \mu_5 \mu_6 \dots \mu_{10}} = -\epsilon^{\mu_6 \dots \mu_{10} \mu_1 \dots \mu_5}$.) So there are only $\frac{256}{2}$ (complex) independent terms in u (and in v).

Next the CG decomposition of a chiral and an antichiral spinor (a spinor and a conjugate spinor) ψ^α and λ_α

$$\psi^\alpha \lambda_\beta = \sum_{k=0,2,4} c_k (\gamma^{\mu_1 \dots \mu_k} C^{-1})^\alpha_\beta (\lambda^T C \gamma_{\mu_1 \dots \mu_k} \psi), \quad (10)$$

The spinor space of $\psi^\alpha \lambda_\beta$ is again 256 dimensional, and the basis of Dirac matrices also has 256 terms

$$1 + \left(\frac{10 \cdot 9}{1 \cdot 2} = 45 \right) + \left(\frac{10 \cdot 9 \cdot 8 \cdot 7}{1 \cdot 2 \cdot 3 \cdot 4} = 210 \right) = 256. \quad (11)$$

Finally we decide on the reality properties of the spinor irrep. The reality property of the spinor irrep is the same as that of the conjugate spinor irrep because they differ only by a sign for

some of the generators, see (3)

$$\frac{1}{2}\gamma^{mn} = \frac{1}{2} \begin{pmatrix} \gamma^{ij}, i\gamma^{(9)} & 0 \\ 0 & \gamma^{ij}, -i\gamma^{(9)} \end{pmatrix} \quad (12)$$

with $i, j = 1, \dots, 9$. If the irrep were real or pseudoreal we should have $(\gamma^{mn})^* = S\gamma^{mn}S^{-1}$. Hence $(i\gamma^j)^* = -i\gamma^j = Si\gamma^jS^{-1}$. Then $S\gamma^j + \gamma^jS = 0$, which would mean that in $d = 9+0$ there is a 10th Dirac matrix. This is not possible, hence the spinor (and conjugate spinor irrep) in $d = 10+0$ is **complex**.